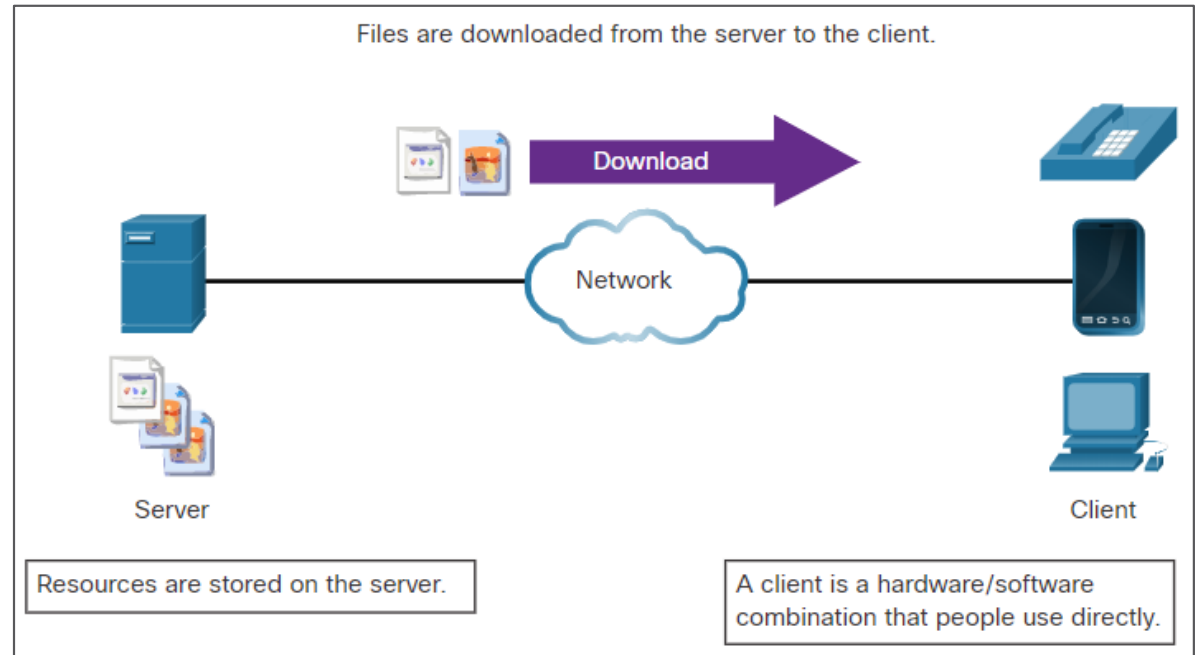


4.3 Linux Servers and Clients

Linux Servers and Clients

An Introduction to Client-Server Communications

- Servers are computers with software installed that enables them to provide services to clients across the network.
- Some provide external resources such as files, email messages, or web pages to clients upon request.
- Other services run maintenance tasks such as log management, disk scanning and so on.
- Each service requires separate server software.
- The server in the figure uses file server software to provide clients with the ability to retrieve and submit files.



Servers, Services, and Their Ports

- A port is a reserved network resource used by a service.
- While the administrator can decide which port to use with any given service, many clients are configured to use a specific port by default.
- The following table lists a few commonly used ports and their services. These are also called as well-known ports.

Port	Description
20/21	File Transfer Protocol (FTP)
22	Secure Shell (SSH)
23	Telnet remote login service
25	Simple Mail Transfer Protocol (SMTP)
53	Domain Name System (DNS)
67/68	Dynamic Host Configuration Protocol (DHCP)

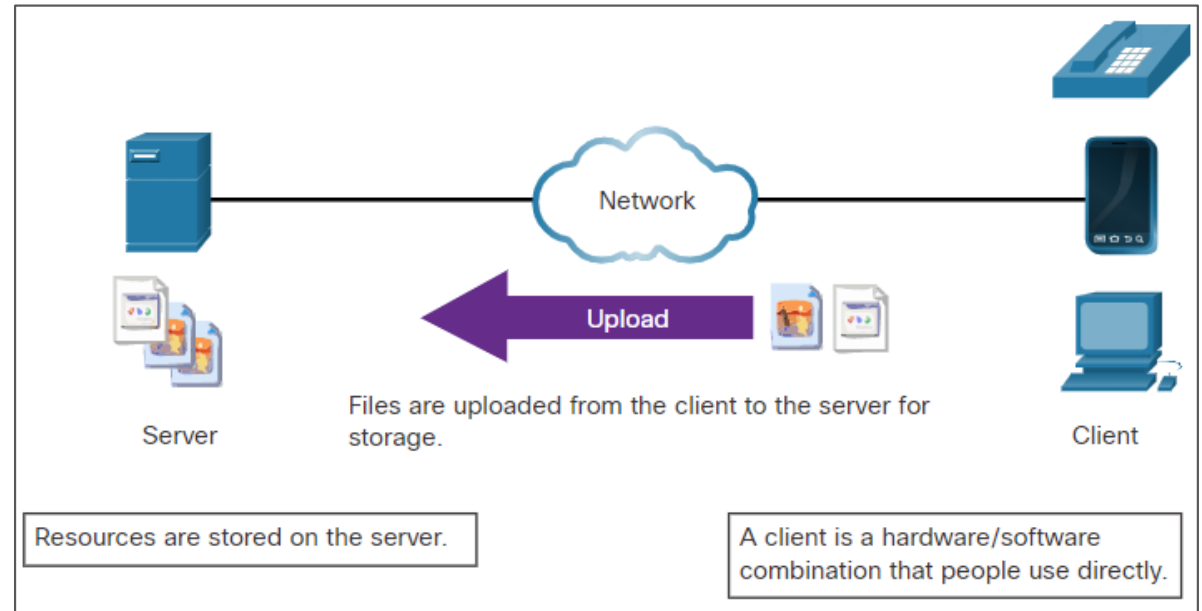
Servers, Services, and Their Ports (Contd.)

Port	Description
69	Trivial File Transfer Protocol (TFTP)
80	Hypertext Transfer Protocol (HTTP)
110	Post Office Protocol version 3 (POP3)
123	Network Time Protocol (NTP)
143	Internet Message Access Protocol (IMAP)
161/162	Simple Network Management Protocol (SNMP)
443	HTTP Secure (HTTPS)

Linux Servers and Clients

Clients

- Clients are programs or applications designed to communicate with a specific type of server.
- Clients use a well-defined protocol to communicate with the server.
- Web browsers are web clients that are used to communicate with web servers through the Hyper Text Transfer Protocol on port 80.
- The File Transfer Protocol client is software used to communicate with an FTP server.
- The figure shows a client uploading files to a server.



Linux Servers and Clients

Lab - Linux Servers

In this lab, you will use the Linux command line to identify servers that are running on a computer.

4.4 Basic Server Administration

Basic Server Administration

Service Configuration Files

- In Linux, services are managed using configuration files.
- Common options in configuration files are port number, location of the hosted resources, and client authorization details.
- When the service starts, it looks for its configuration files, loads them into memory, and adjusts itself according to the settings in the files.
- The command output shows a portion of the configuration file for Nginx, which is a lightweight web server for Linux.

```
[analyst@secOps ~]$ cat /etc/nginx/nginx.conf
#user html;
worker_processes 1;
#error_log logs/error.log;
#error_log logs/error.log notice;
#error_log logs/error.log info;
#pid logs/nginx.pid;
events {
    worker_connections 1024;
}
http {
    include mime.types;
    default_type application/octet-stream;
    #log_format main '$remote_addr - $remote_user [$time_local] "$request" '
    #                '$status $body_bytes_sent "$http_referer" '
    #                '"$http_user_agent" "$http_x_forwarded_for"';
    #access_log logs/access.log main;
```


Basic Server Administration

Service Configuration Files (Contd.)

The command output shows the configuration file for the network time protocol (NTP).

```
[analyst@secOps ~]$ cat /etc/ntp.conf
# Please consider joining the pool:
#
#       http://www.pool.ntp.org/join.html
#
# For additional information see:
# - https://wiki.archlinux.org/index.php/Network_Time_Protocol_daemon
# - http://support.ntp.org/bin/view/Support/GettingStarted
# - the ntp.conf man page
# Associate to Arch's NTP pool
server 0.arch.pool.ntp.org
server 1.arch.pool.ntp.org
server 2.arch.pool.ntp.org
server 3.arch.pool.ntp.org
# By default, the server allows:
# - all queries from the local host
# - only time queries from remote hosts, protected by rate limiting and kod
restrict default kod limited nomodify nopeer noquery notrap
restrict 127.0.0.1
restrict ::1
# Location of drift file
[analyst@secOps ~]$
```

Basic Server Administration

Service Configuration Files (Contd.)

- The command output shows the configuration file for Snort, a Linux-based intrusion detection system (IDS).
- There is no rule for a configuration file format. It is the choice of the service's developer. However, the **option = value** format is often used.

```
[analyst@secOps ~]$ cat /etc/snort/snort.conf
#-----
#   VRT Rule Packages Snort.conf
#
#   For more information visit us at:
#       http://www.snort.org           Snort Website
#       http://vrt-blog.snort.org/     Sourcefire VRT Blog
#
#   Mailing list Contact:  snort-sigs@lists.sourceforge.net
#   False Positive reports: fp@sourcefire.com
#   Snort bugs:           bugs@snort.org
#
#   Compatible with Snort Versions:
#   VERSIONS : 2.9.9.0
#
#   Snort build options:
#   OPTIONS : --enable-gre --enable-mpls --enable-targetbased --enable-ppm --enable-perfprofiling --
enable-zlib --enable-active-response --enable-normalizer --enable-reload --enable-react --enable-
flexresp3
<output omitted>
#####
# Step #1: Set the network variables.  For more information, see README.variables
#####
# Setup the network addresses you are protecting
###ipvar HOME_NET any
###ipvar HOME_NET [192.168.0.0/24,192.168.1.0/24]
ipvar HOME_NET [209.165.200.224/27]
# Set up the external network addresses.  Leave as "any" in most situations
ipvar EXTERNAL_NET any
```

Hardening Devices

- Device hardening involves implementing proven methods of securing the device and protecting its administrative access.
- Some of these methods involve maintaining passwords, configuring enhanced remote login features, and implementing secure login with SSH.
- Depending on the Linux distribution, many services are enabled by default. Stopping such services and ensuring they do not automatically start at boot time is another device hardening technique.
- OS updates are extremely important to maintaining a hardened device. OS developers create and issue fixes and patches regularly.

Hardening Devices (Contd.)

The following are basic best practices for device hardening:

- Ensure physical security
- Minimize installed packages
- Disable unused services
- Use SSH and disable the root account login over SSH
- Keep the system updated
- Disable USB auto-detection
- Enforce strong passwords
- Force periodic password changes
- Keep users from re-using old passwords

Monitoring Service Logs

- Log files are the records that a computer stores to keep track of important events. Kernel, services, and application events are all recorded in log files.
- By monitoring Linux log files, an administrator gains a clear picture of the computer's performance, security status, and any underlying issues.
- In Linux, log files can be categorized as:
 - Application logs
 - Event logs
 - Service logs
 - System logs
- Some logs contain information about daemons that are running in Linux. A daemon is a background process that runs without the need for user interaction.

Monitoring Service Logs (Contd.)

The following table lists a few popular Linux log files and their functions:

Linux Log File	Description
/var/log/messages	• This directory contains generic computer activity logs. • It is mainly used to store informational and non-critical system messages.
/var/log/auth.log	• This file stores all authentication-related events in Debian and Ubuntu computers. • Anything involving the user authorization mechanism can be found in this file.
/var/log/secure	• This directory is used by RedHat and CentOS computers. • It also tracks sudo logins, SSH logins, and other errors logged by SSSD.
/var/log/boot.log	• This file stores boot-related information and messages logged during the computer startup process.

Monitoring Service Logs (Contd.)

Linux Log File	Description
/var/log/dmesg	• This directory contains kernel ring buffer messages. • Information related to hardware devices and their drivers is recorded here. • It is very important because, due to their low-level nature, logging systems such as syslog are not running when these events take place and are unavailable to the administrator in real-time.
/var/log/kern.log	• This file contains information logged by the kernel.
/var/log/cron	• Cron is a service used to schedule automated tasks in Linux and this directory stores its events. • Whenever a scheduled task (or cron job) runs, all its relevant information including execution status and error messages are stored here.
/var/log/mysqld.log or /var/log/mysql.log	• This is the MySQL log file. • All debug, failure and success messages related to the mysqld process and mysqld_safe daemon are logged here.

Basic Server Administration

Monitoring Service Logs (Contd.)

- The command output shows a portion of **/var/log/messages** log file.
- Each line represents a logged event.
- The timestamps at the beginning of the lines mark the moment the event took place.

```
[analyst@secOps ~]$ sudo cat /var/log/messages
Mar 20 15:28:45 secOps kernel: Linux version 4.15.10-1-ARCH (builduser@heftig-18961) (gcc version 7.3.1
20180312 (GCC)) #1 SMP PREEMPT Thu Mar 15 12:24:34 UTC 2018
Mar 20 15:28:45 secOps kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-linux root=UUID=07c6b457-3f39-
4ddf-bfd8-c169e8a877b2 rw quiet
Mar 20 15:28:45 secOps kernel: KERNEL supported cpus:
Mar 20 15:28:45 secOps kernel: Intel GenuineIntel
Mar 20 15:28:45 secOps kernel: AMD AuthenticAMD
Mar 20 15:28:45 secOps kernel: Centaur CentaurHauls
Mar 20 15:28:45 secOps kernel: x86/fpu: Supporting XSAVE feature 0x001: 'x87 floating point registers'
Mar 20 15:28:45 secOps kernel: x86/fpu: Supporting XSAVE feature 0x002: 'SSE registers'
Mar 20 15:28:45 secOps kernel: x86/fpu: Supporting XSAVE feature 0x004: 'AVX registers'
Mar 20 15:28:45 secOps kernel: x86/fpu: xstate_offset[2]: 576, xstate_sizes[2]: 256
Mar 20 15:28:45 secOps kernel: x86/fpu: Enabled xstate features 0x7, context size is 832 bytes, using
'standard' format.
Mar 20 15:28:45 secOps kernel: e820: BIOS-provided physical RAM map:
Mar 20 15:28:45 secOps kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable
Mar 20 15:28:45 secOps kernel: BIOS-e820: [mem 0x000000000009fc00-0x000000000009ffff] reserved
Mar 20 15:28:45 secOps kernel: BIOS-e820: [mem 0x00000000000f0000-0x00000000000fffff] reserved
Mar 20 15:28:45 secOps kernel: BIOS-e820: [mem 0x0000000000100000-0x00000000003fffff] usable
Mar 20 15:28:45 secOps kernel: BIOS-e820: [mem 0x00000000003fff0000-0x00000000003fffff] ACPI data
Mar 20 15:28:45 secOps kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec0ffff] reserved
Mar 20 15:28:45 secOps kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000fee0ffff] reserved
Mar 20 15:28:45 secOps kernel: BIOS-e820: [mem 0x00000000fffc0000-0x00000000ffffffff] reserved
Mar 20 15:28:45 secOps kernel: NX (Execute Disable) protection: active
Mar 20 15:28:45 secOps kernel: random: fast init done
Mar 20 15:28:45 secOps kernel: SMBIOS 2.5 present.
Mar 20 15:28:45 secOps kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/2006
Mar 20 15:28:45 secOps kernel: Hypervisor detected: KVM
Mar 20 15:28:45 secOps kernel: e820: last_pfn = 0x3fff0 max_arch_pfn = 0x400000000
Mar 20 15:28:45 secOps kernel: MTRR: Disabled
Mar 20 15:28:45 secOps kernel: x86/PAT: MTRRs disabled, skipping PAT initialization too.
Mar 20 15:28:45 secOps kernel: CPU MTRRs all blank - virtualized system.
```


Lab – Locating Log Files

In this lab, you will get familiar with locating and manipulating Linux log files.